

CRANBROOK SCHOOL

YEAR 12 MATHEMATICS – 2 UNIT

Term 3 2003

Time : 3 h / GC, CGH, HRK and SKB

All questions are of equal value.

All necessary working should be shown in every question.

Full marks may not be awarded if work is careless or badly arranged.

Approved silent calculators may be used.

Submit your work in ten 4 Page Booklets.

- | | | |
|-----|--|----|
| 1. | (12marks) (Begin a 4 page booklet.) | GC |
| (a) | Calculate $\frac{432 \cdot 5}{18 \cdot 9 \times 4 \cdot 6}$ correct to two decimal places. | 2 |
| (b) | Factorise $3x^2 - x - 10$ | 2 |
| (c) | Solve for x : $3 - \frac{2x}{3} < 4$ | 2 |
| (d) | If $(2\sqrt{3} - 1)(\sqrt{3} + 2) = a + \sqrt{b}$ find the values of a and b . | 2 |
| (e) | Solve and graph on a number line the values of x for which $ x - 2 \leq 8$. | 2 |
| (f) | Express $\frac{3^{-1}a^2}{2a^{-3}}$ with positive indices in its simplest form. | 2 |

2. (12marks) (Begin a 4 page booklet.)

GC

The points A (-3,2) and B (5,8) lie on a number plane.

- | | | |
|-----|--|---|
| (a) | Find the equation of AB in general form. | 2 |
| (b) | Find the midpoint of AB. | 1 |
| (c) | Find the length of AB. | 1 |
| (d) | Show that the equation of the circle with diameter AB is
$x^2 + y^2 - 2x - 10y + 1 = 0$ | 2 |
| (e) | Prove that the point C (1,10) lies on this circle. | 1 |
| (f) | Prove that $AC \perp CB$ | 1 |
| (g) | Prove that the lines $12x - 5y + 3 = 0$ and $24x - 10y - 7 = 0$ are parallel. | 1 |
| (h) | The point A(1, k) lies on the line $12x - 5y + 3 = 0$. Find the value of k . | 1 |
| (i) | Hence find the perpendicular distance between the lines $12x - 5y + 3 = 0$ and $24x - 10y - 7 = 0$. | 2 |

3. (12marks) (Begin a 4 page booklet.)

GC

- (a) Differentiate the following with respect to x .
Leave your answer in its simplest form.

- | | | |
|------|-------------------|---|
| (i) | $(7x^2 - 2)^5$ | 2 |
| (ii) | $\frac{3x}{2x+5}$ | 2 |

- (b) The tangent to the curve $y = 3x^3 - 8x^2$ at the point of contact, P (2,-8), cuts the x -axis at A, and the normal to the curve at the same point of contact cuts the y -axis at B.

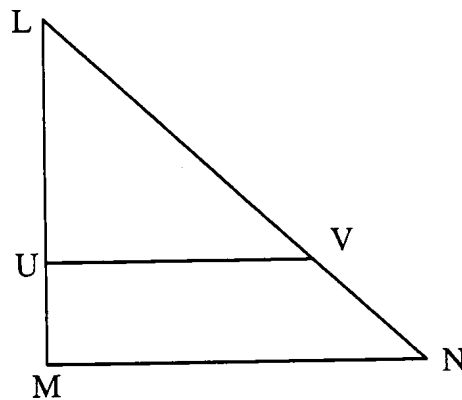
- (i) Find the equation of the tangent at P. 2
- (ii) Find the equation of the normal at P. 2
- (iii) Find the coordinates of A and B. 2

- (c) Find $\lim_{x \rightarrow 1} \frac{3x^2 + x - 4}{x - 1}$ 2

4. (12marks) (Begin a 4 page booklet.)

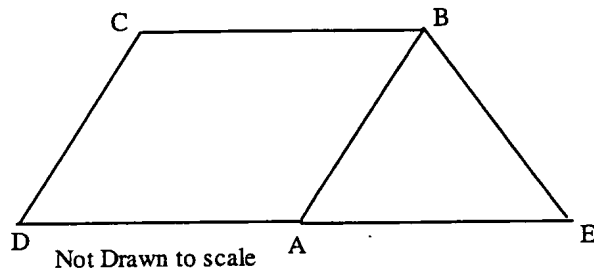
CGH

(a)



- (i) Draw this diagram in your book. 1
- (ii) Consider triangles LMN and LUV where UV is parallel to MN and UM=3m, MN=7m and UV=5m. $LM \perp MN$.
- (A) Find the length of LU 2
- (B) Find the length of VN 1

(b)

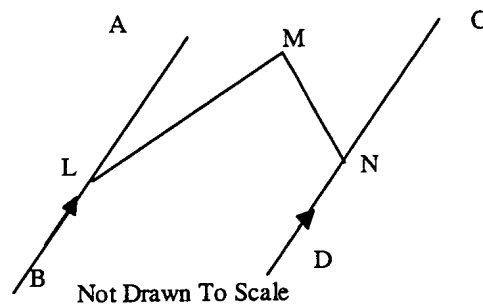


ABCD is a rhombus with $\angle BCD = 120^\circ$ while ABE is an equilateral triangle.

- (i) In your examination booklet draw a neat sketch showing this information.
- (ii) Find the size of $\angle EBC$ giving reasons for your answer.
- (iii) Find the size of $\angle ECB$ giving reasons for your answer.

4

(c)



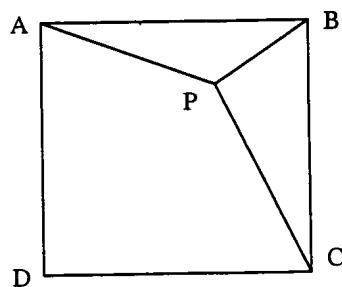
- (i) In your examination booklet draw a neat sketch showing this information.
- (ii) If $\angle MLB = 130^\circ$ and $\angle MND = 110^\circ$ find, giving reasons for your answers:
 - (A) $\angle LMN$
 - (B) $\angle MNC$

4

5. (12marks) (Begin a 4 page booklet.)

CGH

- (a) Sketch the graph of $y = 3 \sin 2x$ for $|x| \leq \pi$ and find the area between the x-axis and the curve $y = 3 \sin 2x$ for $|x| \leq \pi$. 4
- (b) ABCD is a square of side x cm, with P a point within the square such that $PC=6\text{cm}$, $PB=2\text{cm}$ and $AP= 2\sqrt{5}$ cm. Let $\angle PBC = \alpha$.



Not to Scale

- (i) Using the cosine rule in the triangle PBC show that $\cos \alpha = \frac{x^2 - 32}{4x}$. 3
- (ii) By considering triangle PBA, show that $\sin \alpha = \frac{x^2 - 16}{4x}$. 2
- (iii) Hence or otherwise show that the value of x is a solution of $x^4 - 56x^2 + 640 = 0$ 1
- (iv) Find x . Give reasons for your answer. 2

6. (12marks) (Begin a 4 page booklet.)

CGH

(a) If $y = (x - 3)^3$ find all stationary points and determine their nature. 3

(b) If $y = x^3 - 9x$:

- (i) Find any x intercepts. 1
- (ii) Find both $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. 1
- (iii) Determine whether $y = x^3 - 9x$ is an odd or an even function. 1
- (iv) Find any maximum or minimum turning points. 3
- (v) Find any points of inflexion. 2
- (vi) Graph $y = x^3 - 9x$ 1

7. (12marks) (Begin a 4 page booklet.)

HRK

(a) To calculate the area of the region bounded by the curve $y = x^2 - 2x$ and the x axis and between the lines $x = 0$ and $x = 4$, Ernie used $\int_0^4 (x^2 - 2x) dx$.

(i) Explain why Ernie's method of calculating this area is incorrect. 1

(ii) Find the area of the required region. 3

(b) Use Simpson's Rule with 3 function values to approximate the area enclosed between the curve $y = \frac{1}{(x+1)^2}$ and the lines $x = 0$ and $x = 4$ correct to 2 significant figures. 4

(c) Find the volume of the solid formed when the area bounded by the curve $y = 5 - x^2$ for $x \geq 0$, the y axis and the line $y = 1$ is rotated about the y axis. 4

8. (12marks) (Begin a 4 page booklet.)

HRK

(a) For the equation $x^2 + (k+6)x - 2k = 0$ find the:

(i) discriminant in terms of k ; 2

(ii) values of k for which this equation has real roots. 2

(b) Given the equation $3x^2 + 4x - 3 = 0$ has roots α and β , evaluate the following without finding α or β :

(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$ 1

(iii) $2\alpha^2 + 2\beta^2$ 2

(c) A parabola has equation $x^2 + 6x - 33 = 12y$.

(i) Find the coordinates of its vertex. 1

(ii) Find the focal length. 1

(iii) Find the equation of its directrix. 1

(iv) Show that the line $x - 2y + 7 = 0$ is not a focal chord of this parabola. 1

9. (12marks) (Begin a 4 page booklet.)

SKB

(a) Differentiate the following with respect to x
(do not simplify your answer):

(i) $x^3 e^{5x-1}$ 2

(ii) $\log_e \left[\frac{3x^5 - 4}{6x^3 - 5} \right]$ 2

(b) Evaluate the following integral:

$$\int_0^1 \frac{24x^2 - 14}{4x^3 - 7x - 5} dx \quad 4$$

(c) Find the exact area bounded by the curve $y = 2 \ln x$, the x -axis and line $x = 2$. 4

10. (12marks) (Begin a 4 page booklet.)

SKB

- (a) If x , y and 9 are the first three terms of a geometric series and y , x and 2 are the first three terms of an arithmetic sequence, find the values of x and y . 4
- (b) An investor wants to borrow \$1 000 000 to purchase a block of units at Penrith from financial institution X which offers an interest rate of 6% p.a. reducible. The investor is to repay the loan in equal monthly instalments M , over 10 years.
- (i) If A_n is the amount owing after n instalments develop expressions for A_1 , A_2 , A_3 and A_n . Hence show that the monthly instalment, M is given by : $M = \frac{5000(1.005)^{120}}{1.005^{120} - 1}$ 3
- (ii) Calculate the value of the monthly instalment, M to the nearest cent. 1
- (iii) Determine the amount still owing to institution X after 5 years, to the nearest cent. 2
- (iv) If the investor borrows the same amount of money over 10 years from F.B.Knightly Investments Ltd. which offers simple interest at 4.5% p.a., would the investor have been worse off? Explain your answer. 2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

$$\textcircled{1} \text{ (a) } \frac{432.5}{18.9 \times 4.6} = 4.974695... \\ = 4.97 \text{ (2 d.p.)}$$

$$\text{(b) } 3x^2 - x - 10 = (3x+5)(x-2)$$

$$\text{(c) } 3 - \frac{2x}{3} < 4$$

$$\therefore -\frac{2x}{3} < 1$$

$$\therefore x > -1\frac{1}{2}$$

$$\text{(d) } (2\sqrt{3}-1)(\sqrt{3}+2) = a+\sqrt{b}$$

$$\therefore 6+4\sqrt{3}-\sqrt{3}-2 = a+\sqrt{b}$$

$$\therefore 4+3\sqrt{3} = a+\sqrt{b}$$

$$\therefore 4+\sqrt{27} = a+\sqrt{b}$$

$$\therefore a=4, b=27$$

$$\text{(e) } |x-2| \leq 8$$

$$\therefore -8 \leq x-2 \leq 8$$

$$\therefore -6 \leq x \leq 10$$



$$\text{(f) } \frac{3^{-1}a^2}{2a^{-3}} = \frac{a^5}{6}$$

$$\textcircled{2} \quad A=(-3,2), B=(5,8)$$

$$\text{(a) Eqn of AB is: } \frac{y-2}{x+3} = \frac{8-2}{5+3}$$

$$8(y-2) = 6(x+3)$$

$$8y - 16 = 6x + 18$$

$$6x - 8y + 34 = 0$$

$$\therefore 3x - 4y + 17 = 0$$

$$\text{(b) } M_{AB} = \left(\frac{-3+5}{2}, \frac{2+8}{2} \right) \\ = (1, 5)$$

$$\text{(c) } \text{length AB} = \sqrt{(5-(-3))^2 + (8-2)^2} \\ = \sqrt{64+36} \\ = 10 \text{ units.}$$

(d) If AB is the diameter of the circle $\Rightarrow$ centre = (1, 5)
radius = 5 units

$$\therefore \text{Eqn of circle is: } (x-1)^2 + (y-5)^2 = 5^2$$

$$x^2 - 2x + 1 + y^2 - 10y + 25 = 25$$

$$\therefore x^2 + y^2 - 2x - 10y + 1 = 0$$

$$\text{(e) Sub } (1,10) \text{ into } x^2 + y^2 - 2x - 10y + 1 = 0$$

$$\therefore \text{LHS} = 1 + 100 - 2 - 100 + 1$$

$$= 0$$

$$= \text{RHS}$$

$\Rightarrow C(1,10)$ lies on the given circle

$$\text{(f) } m_{AC} = \frac{10-2}{1-3} = 2$$

$$m_{CB} = \frac{10-8}{1-5} = -\frac{1}{2}$$

$$\text{AS } m_{AC} \cdot m_{CB} = -1$$

$$\Rightarrow AC \perp CB$$

$$\text{(g) For } 12x - 5y + 3 = 0 \quad m_1 = \frac{-12}{-5} \\ = \frac{12}{5}$$

$$\text{For } 24x - 10y - 7 = 0 \quad m_2 = \frac{-24}{-10} \\ = \frac{12}{5}$$

As $m_1 = m_2 \Rightarrow$ lines are parallel

$$\text{(h) Sub } (1,k) \text{ into } 12x - 5y + 3 = 0$$

$$\therefore 12 - 5k + 3 = 0$$

$$\therefore 5k = 15$$

$$\therefore k = 3$$

$$\text{(i) perp. distance} = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$\text{where } (x_1, y_1) = (1, 3); (A, B, C) = (24, -10, -7)$$

$$\therefore \text{perp. distance} = \frac{|24 - 30 - 7|}{\sqrt{24^2 + (-10)^2}} = \frac{|-13|}{26} = \frac{1}{2}$$

(3) (a)(i) Let $y = (7x^2 - 2)^5$
 $\therefore \frac{dy}{dx} = 5(7x^2 - 2)^4 \cdot 14x$
 $= 70x(7x^2 - 2)^4$

(ii) Let $y = \frac{3x}{2x+5}$

$$\therefore \frac{dy}{dx} = \frac{(2x+5) \cdot 3 - 3x \cdot 2}{(2x+5)^2}$$

$$= \frac{6x+15-6x}{(2x+5)^2}$$

$$= \frac{15}{(2x+5)^2}$$

(b) $y = 3x^3 - 8x^2$

(i) $\frac{dy}{dx} = 9x^2 - 16x$

At $P(2, -8)$ $\frac{dy}{dx} = 4 = m_{\text{tangent}}$

$\therefore$ Eqn of tangent at P is:

$$y - (-8) = 4(x - 2)$$

$$\therefore y + 8 = 4x - 8$$

$$\therefore \underline{4x - y - 16 = 0}$$

(ii) At $P(2, -8)$ $m_{\text{normal}} = -\frac{1}{4}$

$\therefore$ Eqn of normal at P is:

$$y - (-8) = -\frac{1}{4}(x - 2)$$

$$\therefore 4y + 32 = -x + 2$$

$$\therefore \underline{x + 4y + 30 = 0}$$

(iii) At A tangent cuts x-axis

$$\therefore y = 0 \quad \therefore 4x - 16 = 0$$

$$\therefore x = 4$$

$$\Rightarrow A = (4, 0)$$

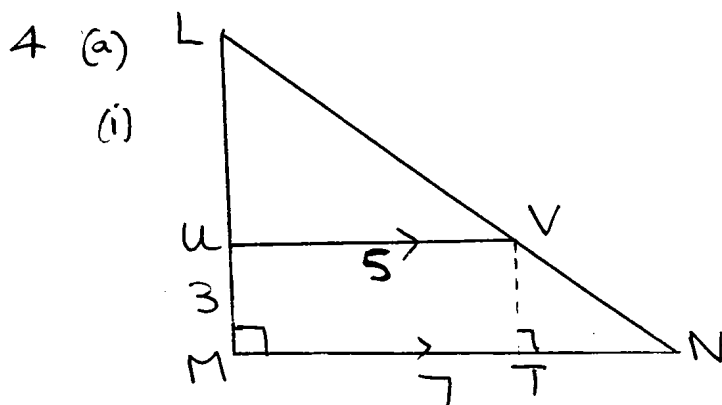
At B normal cuts y-axis

$$\therefore x = 0 \quad \therefore 4y + 30 = 0$$

$$\therefore y = -7\frac{1}{2}$$

$$\Rightarrow B = (0, -7\frac{1}{2})$$

(c) $\lim_{x \rightarrow 1} \frac{5x^2 + 2x - 7}{x - 1}$
 $= \lim_{x \rightarrow 1} \frac{(3x+4)(x-1)}{(x-1)} \quad (x \neq 1)$
 $= \lim_{x \rightarrow 1} (3x+4)$
 $= 7$



(ii) (A) In Δ s LUV and LMN
 $\angle LUV = \angle LMN$ (corr. $\angle$ s in $\parallel$ lines are equal)

Similarly $\angle LVU = \angle LNM$

$\angle L$ is common

$\therefore \Delta LUV \sim \Delta LMN$ (As are equiangular)

$$\therefore \frac{LU}{LU+3} = \frac{5}{7} \quad \left(\begin{array}{l} \text{corr. sides} \\ \text{of similar } \Delta \text{s} \\ \text{are in the} \\ \text{same ratio.} \end{array} \right)$$

$$\therefore 7LU = 5LU + 15$$

$$\therefore 2LU = 15$$

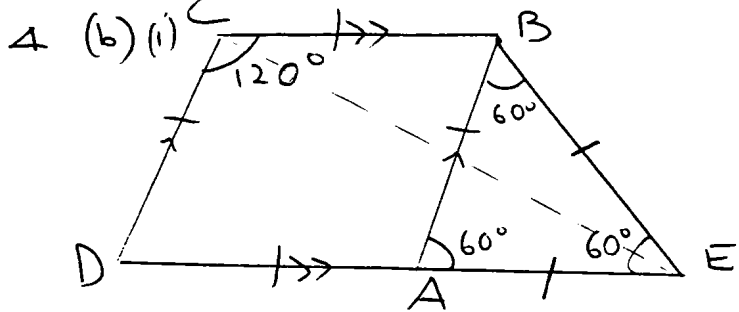
$$\therefore LU = 7.5 \text{ m}$$

(B) Construct $VT \perp MN$

$$\therefore VT = 3, TN = 2$$

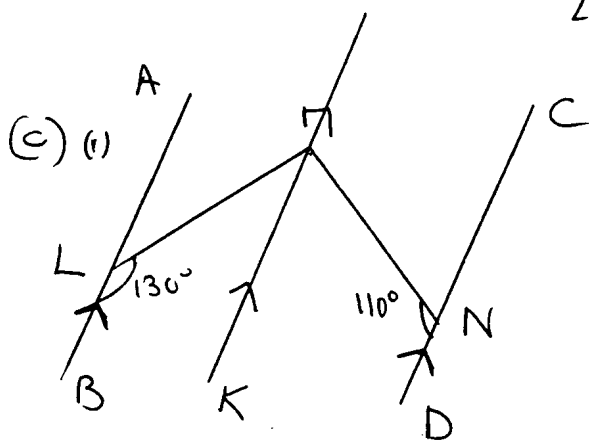
$$\therefore VN = \sqrt{3^2 + 2^2}$$

$$= \sqrt{13} \text{ m}$$



(ii) $DC \parallel BA$ (property of rhombus)
 $\angle CBA = 60^\circ$ (co-int. $\angle$ s in $\parallel$ lines are supp.)
 $\angle ABE = 60^\circ$ (property of equil. Δ)
 $\therefore \angle EBC = 120^\circ$ ($\angle$ sum of adj. $\angle$ s)

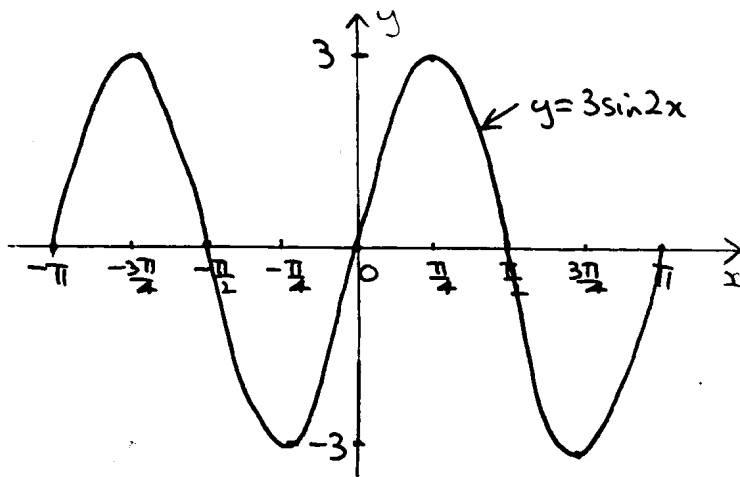
(iii) As $CB = AB$ (property of rhombus)
 and $AB = BE$ (property of equil. Δ)
 $\therefore CB = BE$
 $\Rightarrow \Delta CBE$ is isosceles.
 $\therefore \angle ECB = \frac{180^\circ - 120^\circ}{2}$
 $= 30^\circ$ (base $\angle$ s of isos. Δ are equal;
 $\angle$ sum of $\Delta = 180^\circ$)



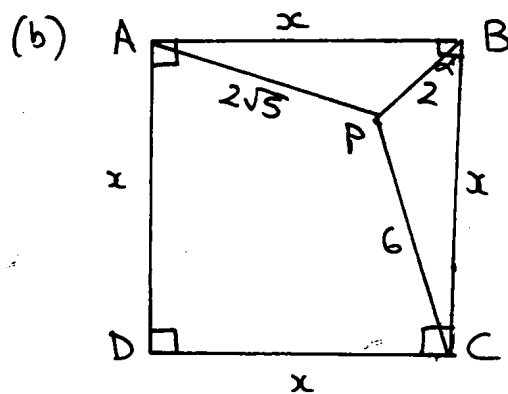
(ii) (A) Construct $MK \parallel BA \parallel DC$
 $\therefore \angle LMK = 50^\circ$ (co-int. $\angle$ s in $\parallel$ lines are supp.)
 Similarly $\angle KMN = 70^\circ$
 $\therefore \angle LMN = 120^\circ$ ($\angle$ sum of adj. $\angle$ s)

(B) $\angle MNC = 70^\circ$ ($\angle$ sum on st. line $= 180^\circ$)

5. (a) $y = 3 \sin 2x$, $|x| \leq \pi$
 Period $= \frac{2\pi}{2} = \pi$; Amplitude $= 3$ units



Required Area $= 4 \int_0^{\frac{\pi}{2}} 3 \sin 2x \, dx$
 $= 12 \left[-\frac{\cos 2x}{2} \right]_0^{\frac{\pi}{2}}$
 $= 6 [-\cos \pi + \cos 0]$
 $= 12 \text{ units}^2$



(i) In ΔPBC : $\cos \angle C = \frac{2^2 + x^2 - 6^2}{2 \times 2 \times x}$
 $= \frac{x^2 - 32}{4x}$ — (1)

(ii) In ΔPBA $\angle ABP = (\frac{\pi}{2} - \angle C)$
 $\therefore \cos(\frac{\pi}{2} - \angle C) = \frac{x^2 + 2^2 - (2\sqrt{5})^2}{2 \times 2 \times x}$
 $= \frac{x^2 - 16}{4x} = \sin \angle C$ — (2)

(ii) Now as $\sin^2 \alpha + \cos^2 \alpha = 1$

sub. ① and ②: $\left(\frac{x^2-16}{4x}\right)^2 + \left(\frac{x^2-32}{4x}\right)^2 = 1$

$\therefore x^4 - 32x^2 + 256 + x^4 - 64x^2 + 1024 = 16x^2$

$\therefore 2x^4 - 112x^2 + 1280 = 0$

$\therefore x^4 - 56x^2 + 640 = 0$

(iv)
$$x^2 = \frac{56 \pm \sqrt{(-56)^2 - 4 \cdot 1 \cdot 640}}{2}$$

$$= \frac{56 \pm 24}{2}$$

$$= 40 \text{ or } 16$$

$\therefore x = 2\sqrt{10}$ ($x^2 > 32$ otherwise
 ① would not yield
 an acute value for $\angle$)

6 (a) $y = (x-3)^3$

$\frac{dy}{dx} = 3(x-3)^2, \frac{d^2y}{dx^2} = 6(x-3)$

For a stat. pt $\frac{dy}{dx} = 0 \therefore x = 3$

x	3 ⁻	3	3 ⁺
$\frac{d^2y}{dx^2}$	+	0	+

$\rightarrow \frac{d^2y}{dx^2} > 0 \Rightarrow$ horiz.
 pt. of inflexion
 on a rising
 curve at (3,0)

$\therefore$ the only stationary point is a
 horizontal point of inflexion at (3,0).

(b) $y = x^3 - 9x$

(i) For x-intercepts $y = 0$

$\therefore 0 = x(x^2 - 9) \therefore x = 0, \pm 3$

(ii) $\frac{dy}{dx} = 3x^2 - 9, \frac{d^2y}{dx^2} = 6x$

(iii) Let $y = f(x) = x^3 - 9x$

Now $f(-x) = -x^3 + 9x = -f(x)$

$\Rightarrow y = x^3 - 9x$ is an odd function.

(iv) For a stat. pt $\frac{dy}{dx} = 0$

$\therefore 3x^2 - 9 = 0 \therefore x = \pm \sqrt{3}$

when $x = \sqrt{3} \frac{dy}{dx} > 0 \Rightarrow$ min. turning
 pt at $(\sqrt{3}, -6\sqrt{3})$

As the function is odd $\Rightarrow$ max. turning pt
 at $(-\sqrt{3}, 6\sqrt{3})$

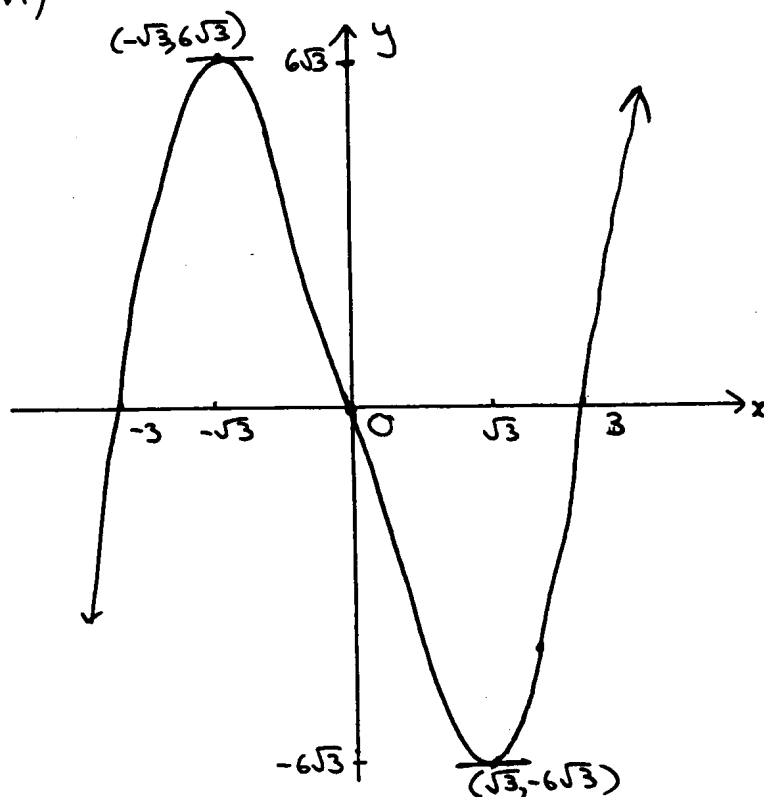
(v) For a possible pt of inflexion $\frac{d^2y}{dx^2} = 0$

$\therefore 6x = 0 \therefore x = 0$

x	0 ⁻	0	0 ⁺
$\frac{d^2y}{dx^2}$	-	0	+

concavity change $\Rightarrow$ pt. of inflexion
 at (0,0).

(vi)



(b) $y = x^3 - 9x$

(i) For x-intercepts $y = 0$

$\therefore 0 = x(x^2 - 9) \therefore x = 0, \pm 3$

(ii) $\frac{dy}{dx} = 3x^2 - 9, \frac{d^2y}{dx^2} = 6x$

(iii) Let $y = f(x) = x^3 - 9x$

Now $f(-x) = -x^3 + 9x = -f(x)$

$\Rightarrow y = x^3 - 9x$ is an odd function.

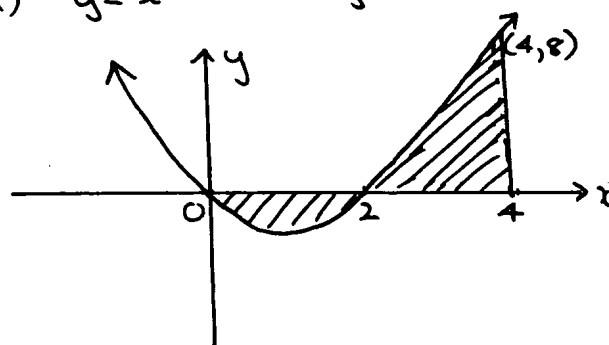
(iv) For a stat. pt $\frac{dy}{dx} = 0$

$\therefore 3x^2 - 9 = 0 \therefore x = \pm \sqrt{3}$

when $x = \sqrt{3} \frac{dy}{dx} > 0 \Rightarrow$ min. turning
 pt at $(\sqrt{3}, -6\sqrt{3})$

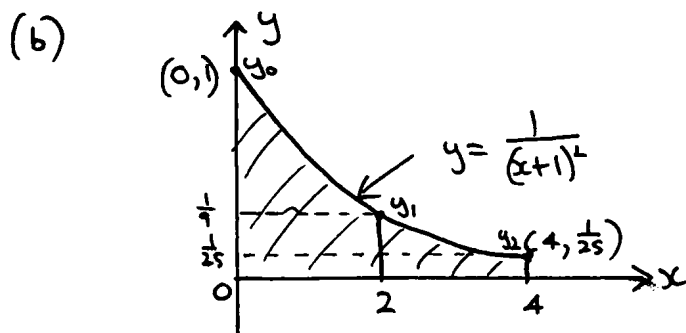
7 (a) $y = x^2 - 2x \therefore y = x(x-2)$

(i)



Ernie's method to find the area
 was wrong as $\int_0^4 (x^2 - 2x) dx$
 did not take into account that
 part of the area was below the
 x-axis.

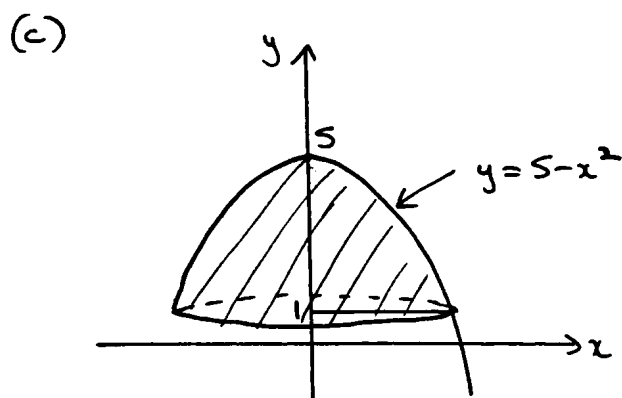
$$\begin{aligned}
 \text{(ii) Req'd area} &= \left| \int_0^2 (x^2 - 2x) dx \right| + \left| \int_2^4 (x^2 - 2x) dx \right| \\
 &= \left| \left[\frac{x^3}{3} - x^2 \right]_0^2 \right| + \left| \left[\frac{x^3}{3} - x^2 \right]_2^4 \right| \\
 &= \left| \left(\frac{8}{3} - 4 \right) - 0 \right| + \left| \left(\frac{64}{3} - 16 \right) - \left(\frac{8}{3} - 4 \right) \right| \\
 &= 8 \text{ units}^2
 \end{aligned}$$



By Simpson's Rule, $\int_0^4 \frac{1}{(x+1)^2} dx \approx \frac{h}{3} [y_0 + y_2 + 4y_1]$

$$h = \frac{4-0}{2} = 2$$

$$\begin{aligned}
 \therefore \text{Area} &= \frac{2}{3} \left[1 + \frac{1}{25} + 4\left(\frac{1}{9}\right) \right] \\
 &= \frac{668}{675} \\
 &= 0.99 \text{ units}^2 \text{ (2 sig. figs)}
 \end{aligned}$$



$$\begin{aligned}
 \text{Volume} &= \pi \int_{-2}^2 x^2 dy \\
 &= \pi \int_{-2}^2 (5-y) dy \\
 &= \pi \left[\frac{(5-y)^2}{-2} \right]_{-2}^2 \\
 &= -\frac{\pi}{2} [0 - 16] \\
 &= 8\pi \text{ units}^3
 \end{aligned}$$

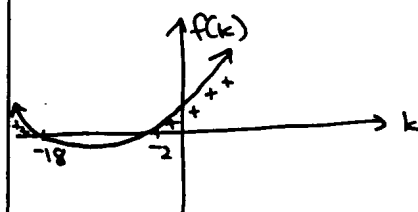
(8) (a) $x^2 + (k+6)x - 2k = 0$

$$\begin{aligned}
 \text{(i) } \Delta &= b^2 - 4ac \\
 &= (k+6)^2 - 4 \cdot 1 \cdot (-2k) \\
 &= k^2 + 12k + 36 + 8k \\
 &= k^2 + 20k + 36
 \end{aligned}$$

(ii) For real roots $\Delta \geq 0$

$$\therefore k^2 + 20k + 36 \geq 0$$

$$\therefore (k+18)(k+2) \geq 0$$



$$\Rightarrow k \leq -18 \text{ or } k \geq -2$$

(b) $3x^2 + 4x - 3 = 0$

(i) $\alpha + \beta = -\frac{b}{a} = -\frac{4}{3}$

(ii) $\alpha\beta = \frac{c}{a} = -\frac{3}{3} = -1$

$$\begin{aligned}
 \text{(iii) } 2\alpha^2 + 2\beta^2 &= 2(\alpha^2 + \beta^2) \\
 &= 2((\alpha + \beta)^2 - 2\alpha\beta) \\
 &= 2\left(\left(-\frac{4}{3}\right)^2 + 2\right) \\
 &= 7\frac{5}{9}
 \end{aligned}$$

(c) $x^2 + 6x - 33 = 12y$

$$\therefore (x+3)^2 - 9 - 33 = 12y$$

$$\therefore (x+3)^2 = 12y + 42$$

$$\therefore (x+3)^2 = 12\left(y + 3\frac{1}{2}\right)$$

$$\text{is of form } (x-h)^2 = 4a(y-k)$$

(i) vertex $x = (-3, -3\frac{1}{2})$

(ii) focal length = 3 units

(iii) Directrix is: $y = -6\frac{1}{2}$

(iv) Now focus is: $(-3, -\frac{1}{2})$

$$\therefore \text{sub } (-3, -\frac{1}{2}) \text{ into } x - 2y + 7 = 0$$

$$\text{LHS} = -3 + 1 + 7$$

$$= 5$$

$$\neq 0$$

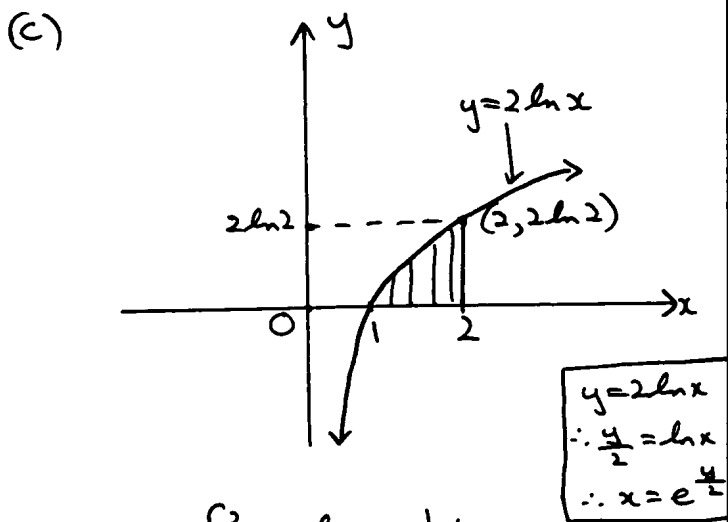
$$\neq \text{RHS}$$

$$\therefore x - 2y + 7 = 0 \text{ is not a focal chord of the parabola.}$$

9(a) (i) let $y = x^3 e^{5x-1}$
 $\therefore \frac{dy}{dx} = x^3 \cdot 5e^{5x-1} + e^{5x-1} \cdot 3x^2$

(ii) let $y = \log_e \left[\frac{3x^5 - 4}{6x^3 - 5} \right]$
 $\therefore y = \log_e [3x^5 - 4] - \log_e [6x^3 - 5]$
 $\therefore \frac{dy}{dx} = \frac{15x^4}{3x^5 - 4} - \frac{18x^2}{6x^3 - 5}$

(b) $I = \int_0^1 \frac{24x^2 - 14}{4x^3 - 7x - 5} dx$
 $= \int_0^1 \frac{2(12x^2 - 7)}{4x^3 - 7x - 5} dx$
 $= 2 \left[\ln |4x^3 - 7x - 5| \right]_0^1$
 $= 2 [\ln |-8| - \ln |-5|]$
 $= 2 \ln \frac{8}{5}$



Area = $\int_1^2 2 \ln x dx$
 $= \text{Area rectangle} - \int_0^{2 \ln 2} x dy$
 $= 2 \times 2 \ln 2 - \int_0^{2 \ln 2} e^{y/2} dy$
 $= 4 \ln 2 - \left[\frac{e^{y/2}}{1/2} \right]_0^{2 \ln 2}$
 $= 4 \ln 2 - 2 [e^{\ln 2} - e^0]$
 $= 4 \ln 2 - 2 [2 - 1]$
 $= (4 \ln 2 - 2) \text{ units}^2$

10(a) If x, y and q are in A.P.
 $\therefore \frac{y}{x} = \frac{q}{y} \therefore y^2 = qx \text{ --- (1)}$

If y, x and 2 are in A.P.
 $\therefore x - y = 2 - x \therefore 2x - 2 = y \text{ --- (2)}$

sub (2) into (1): $(2x - 2)^2 = qx$
 $\therefore 4x^2 - 8x + 4 = qx$
 $\therefore 4x^2 - 17x + 4 = 0$
 $(4x - 1)(x - 4) = 0$
 $\therefore x = \frac{1}{4} \text{ or } 4$

when $x = \frac{1}{4} \quad y = -1\frac{1}{2}$
 $x = 4 \quad y = 6$

$\Rightarrow (x, y) = (\frac{1}{4}, -1\frac{1}{2}) \text{ or } (4, 6)$

(b) (i) 6% p.a. = $\frac{6}{12} \% \text{ p. month}$
 $= 0.005 \text{ p. month}$

Let A_n be the amount owing after n instalments
 Let $P = 1,000,000$
 After 1 instalment amount owing, $A_1 = P(1.005) - \Gamma$
 " 2 instalments " " , $A_2 = A_1(1.005) - \Gamma$
 $= (P(1.005) - \Gamma)(1.005) - \Gamma$
 $= P(1.005)^2 - \Gamma(1 + 1.005)$
 " 3 instalments, " " , $A_3 = A_2(1.005) - \Gamma$
 $= (P(1.005)^2 - \Gamma(1 + 1.005))(1.005) - \Gamma$

$\therefore A_3 = P(1.005)^3 - \Gamma(1.005^2 + 1.005 + 1)$

$\therefore$ continuing this process the amount owing after n instalments, $A_n = P(1.005)^n - \Gamma(1.005^{n-1} + \dots + 1.005 + 1)$

Now after n instalments, $A_n = 0$.

After 10 years $n = 10 \times 12 = 120$

$\therefore 0 = 1,000,000(1.005)^{120} - \Gamma(1 + 1.005 + \dots + 1.005^{119})$

$\therefore \Gamma = \frac{1,000,000(1.005)^{120}}{1 + 1.005 + \dots + 1.005^{119}}$
 A.P. $a=1, r=1.005, n=120$

$\therefore \Gamma = \frac{1,000,000(1.005)^{120}}{1 \left[\frac{1.005^{120} - 1}{0.005} \right]}$

$\therefore \Gamma = \frac{5000(1.005)^{120}}{1.005^{120} - 1}$

(ii) $\therefore$ Monthly instalment, $\Gamma = \$11102.05$
(to the nearest cent)

(iii) After 5 years, $n = 5 \times 12 = 60$.

$$\therefore \text{Amount owing, } A_{60} = 1000000(1.005)^{60} - 11102.05 [1 + 1.005 + \dots + 1.005^{59}]$$

$$\therefore A_{60} = 1000000(1.005)^{60} - 11102.05 \left[\frac{1.005^{60} - 1}{0.005} \right]$$
$$= \$574259.79 \text{ (to nearest cent)}$$

(iv) Interest paid over 10 years

$$= 120 \times \$11102.05 - 1000000$$

$$= \$332246$$

Now Simple Interest = $\frac{P \cdot R \cdot T}{100}$
@ 4.5% p.a.
over 10 years

$$= \frac{1000000 \times 4.5 \times 10}{100}$$
$$= \$450000$$

$\therefore$ The investor would have been worse off by some \$117754 if he had chosen to borrow from F.B. Knightly Investments Ltd. instead of financial institution X.